

Lab Project

List of feasible projects

1. **DAC:** Implement an 8-bit digital to analog converter. Apply different binary codes and find out what analog levels they generate. Is the characteristics linear? Quantify the non-linearity in the characteristics.
2. **Notch filter:** Design and implement a band-stop (or notch) filter. The filter should not allow 50 hertz to pass through, but all other frequencies should be able to pass. Formalize the characteristics of your filter. Can it really pass all other frequencies?
3. **Oscillator:** Design and implement a sine-wave generator. Can the oscillator output amplitude and frequency also be controlled?
4. **Low-pass filters:** For this you will need a line-out cable from your cell phone/laptop (plugs into the audio jack). You have to figure out the terminals of the line-out cable, connect to a bread-board. Now make a tunable R-C low-pass filter. Also, you need to make a power amplifier. Connect the output to a speaker. Demonstrate. Find out how music sounds when the cut-off of the filter is changed. Also, work out the same project with a high-pass filter.
5. **Open-door alarm:** Design and implement an open-door alarm.
6. **Motor speed control:** A 12 volt 0.5 ampere permanent magnet DC motor is to be used. Could you design a circuit to control the speed of the motor? (One possibility is to use a 555-timer to create a pulsating supply to the DC motor and control its speed. The timer cannot supply power to the motor by itself, you might need to use a transistor and a diode for this purpose.)
7. **Function generator:** In this project, build a function generator that can generate triangle and square waves for a variety of frequencies. Use the potentiometers to vary the signal amplitude and the frequency.
8. **Pull pin alarm system:** Implement an alarm circuit to thwart snatching of your valuables while travelling. The alarm circuit kept in your bag sounds a loud alarm, if someone attempts to snatch your bag. This will draw the attention of other passengers, and the burglar can be caught red-handed. In the standby mode, the circuit is locked by a plug and socket arrangement (a mono plug with shorted leads plugged into the mono-jack socket of the unit). When the burglar tries to snatch the bag, the plug detaches from the unit's socket and activates the alarm.
9. **Fridge door alarm:** Implement a light sensing-based door alarm circuit. The circuit should be placed in the fridge (enclosed in a small box) near the lamp or close to the opening. When the door is opened, the light from the lamp or outside will fall on it, and the buzzer will go off for a certain time and stop. This process repeats until the door is closed. This circuit is useful when the fridge door is not closed or it is kept open for a long time.